



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : BACHELOR OF COMPUTER ENGINEERING TECHNOLOGY (COMPUTER SYSTEMS) WITH HONOURS

Course Code & Name : IPB 49906 FINAL YEAR PROJECT 2

Course Lecturer(s) : Ts. DIYANA BT AB KADIR (FYP2 BCE COORDINATOR)

Submission Title : PROGRESS REPORT

Deadline : Day 29 Month Mar Year 2026 Time 4PM

Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

☒	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
4/8/2026 9:59 PM	MD EJAJ MAHUD	5222222123

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
4/8/2026 9:59 PM	IPB 49906	Progress Report 1	5222222123



**UNIVERSITI KUALA LUMPUR
RUBRIC FOR PROJECT PROGRESS**

COURSE CODE	IPB 49906	TOTAL MARKS (50 MARKS)
COURSE NAME	FINAL YEAR PROJECT 2	
COURSE SEMESTER/ YEAR	MARCH 2026	
STUDENT NAME	MD EAJ MAHMUD	
STUDENT ID	52222222123	
ASSESSMENT DATE	WEEK 1 – WEEK 11	Smart Recycling: Reward-Based Reverse Vending Machine Using Proximity Sensor Detection for Plastic Waste Management
ASSESSMENT DUE	29 MAY 2026 4PM	
NAME OF SUPERVISOR	Dr. Grad. Engr. Hannah Binti Sofian	

NO.	CRITERIA	Description	WEIGHTAGE	SCORE 1-10	ACTUAL SCORE
1	Consultation with Supervisor	Meet Supervisor with significant progress	1		
2	Timelines	Record of research progress with respect to the Timelines	1		
3	Research progress	Show progress in research with related information and resources	1		
4	Proposed Idea	Ability to apply higher order thinking	1		
5	Self-driven & Independency	Ability to perform independently	1		
TOTAL SCORE (50 MARKS)					



PROGRESS REPORT NO. 1

Student's Name: MD EJAJ MAHMUD

FYP Code : IPB49906

Semester	07	Week	02
Meeting No.	01	Date	08/04/2026

Progress (fill in by Student)

During Week 1 and Week 2, I focused on checking the hardware requirements for my project before starting the prototype work. I went through each component one by one and reviewed its function, working voltage, compatibility with the Arduino Mega, expected use in the circuit, and current online price. This helped me confirm which parts were really needed and avoid ordering unsuitable items.

After completing that review, I placed online orders for the main components required for the first stage of development. At this stage, the project progress is mainly on preparation and purchasing, but this is an important step because the assembly, wiring, and testing can only begin after the components arrive.

From this process, I also got a clearer picture of the estimated hardware cost and how the selected sensors, display, servo motors, and supporting parts will fit into the overall system design. Once the ordered items are delivered, I plan to start the initial hardware checking, wiring setup, and basic sensor testing.

Appendix 1 shows the ordered items list for Progress Report 1.

Products Ordered

[Robotedu.my Malaysia Robotics Store](#)

Female to Female (FF) 40pcs Dupont Jumper W...

Female to Male (FM) 40pcs Dupont Jumper Way...

MG995 MG996r SG90 9g MG90s S3003 Metal / ...

12V 2A AC to DC Power Supply CCTV Alarm Ard...

Product Care Service Programme New
Protect your purchase with coverage against damage, fire, flood
enjoy added value like discounts on Home Improvement, Recyc
as part of Service Programme. [Learn more](#)

Mega 2560 Atmel ATMEGA2560 / CH340G / PR...

Ultrasonic Sensor HC-SR04 HC SR 04 HCSR04 U...

Male to Male (MM) 40pcs Dupont Jumper Waya...

LCD 1602 / 2004 16x2 20x4 LCD Screen Liquid ...

MB102 Solderless Mini Medium Large Breadboa...

Adjustable IR Infrared Range Finder Obstacles A...

LM2596 5V 9V 12V 24V DC to DC DC-DC Step D...

【10PCS】 LED LEDs 3MM 5MM 10MM Light E...

【10PCS】 LED LEDs 3MM 5MM 10MM Light E...

【10PCS】 LED LEDs 3MM 5MM 10MM Light E...

Electronic Components - 5V Piezo Buzzer with ...

Solderless 2.1mm Female Terminal Block DC Ja...

Resistor 220 ohm (Carbon - 0.25W)

Resistor 1k ohm (Carbon - 0.25W)

Resistor 10k ohm (Carbon - 0.25W)

LJ18A3-B-Z/BX LJ18A3 NPN M18 6-36VDC 1...

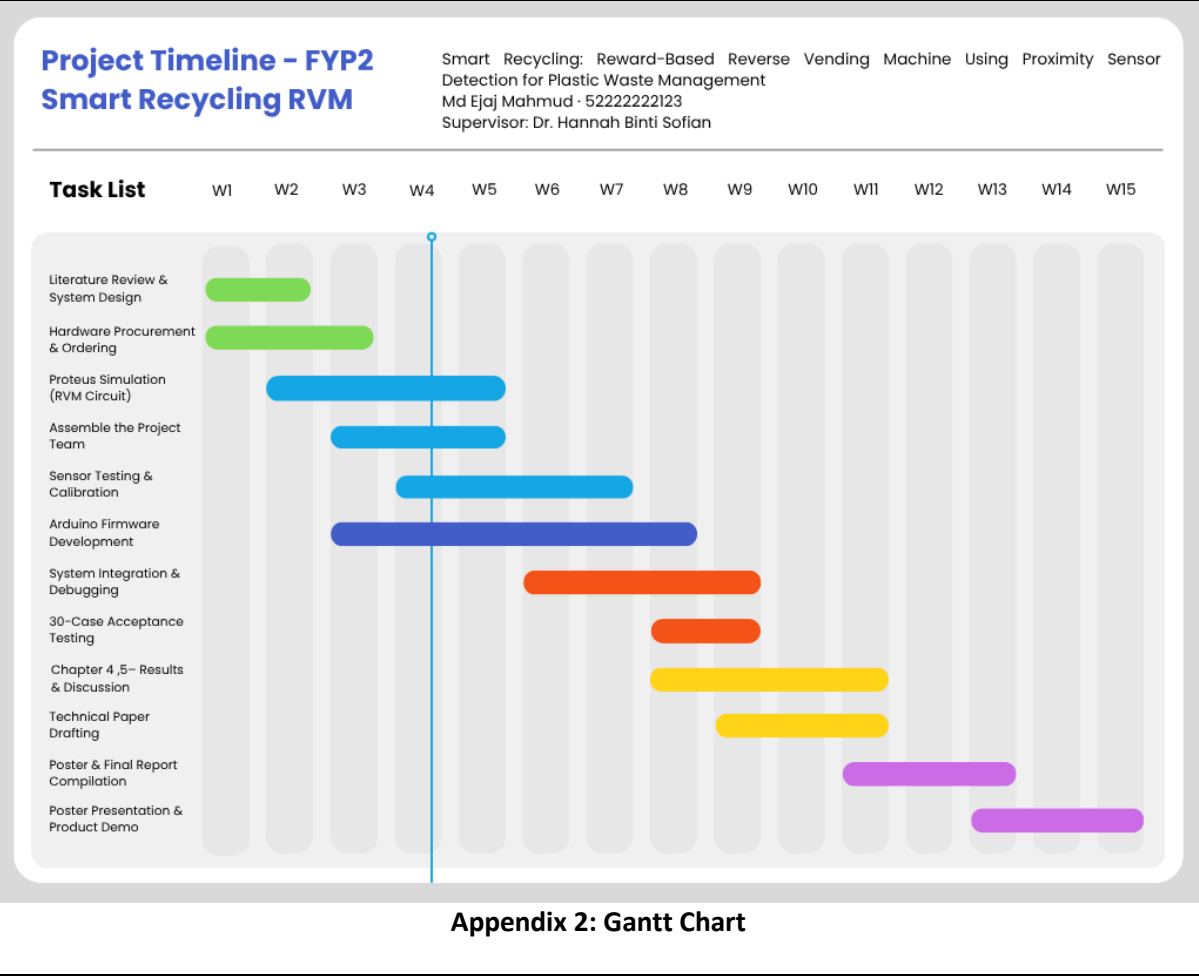
Mega 2560 R3 Compatible with USB Cable 16U2...

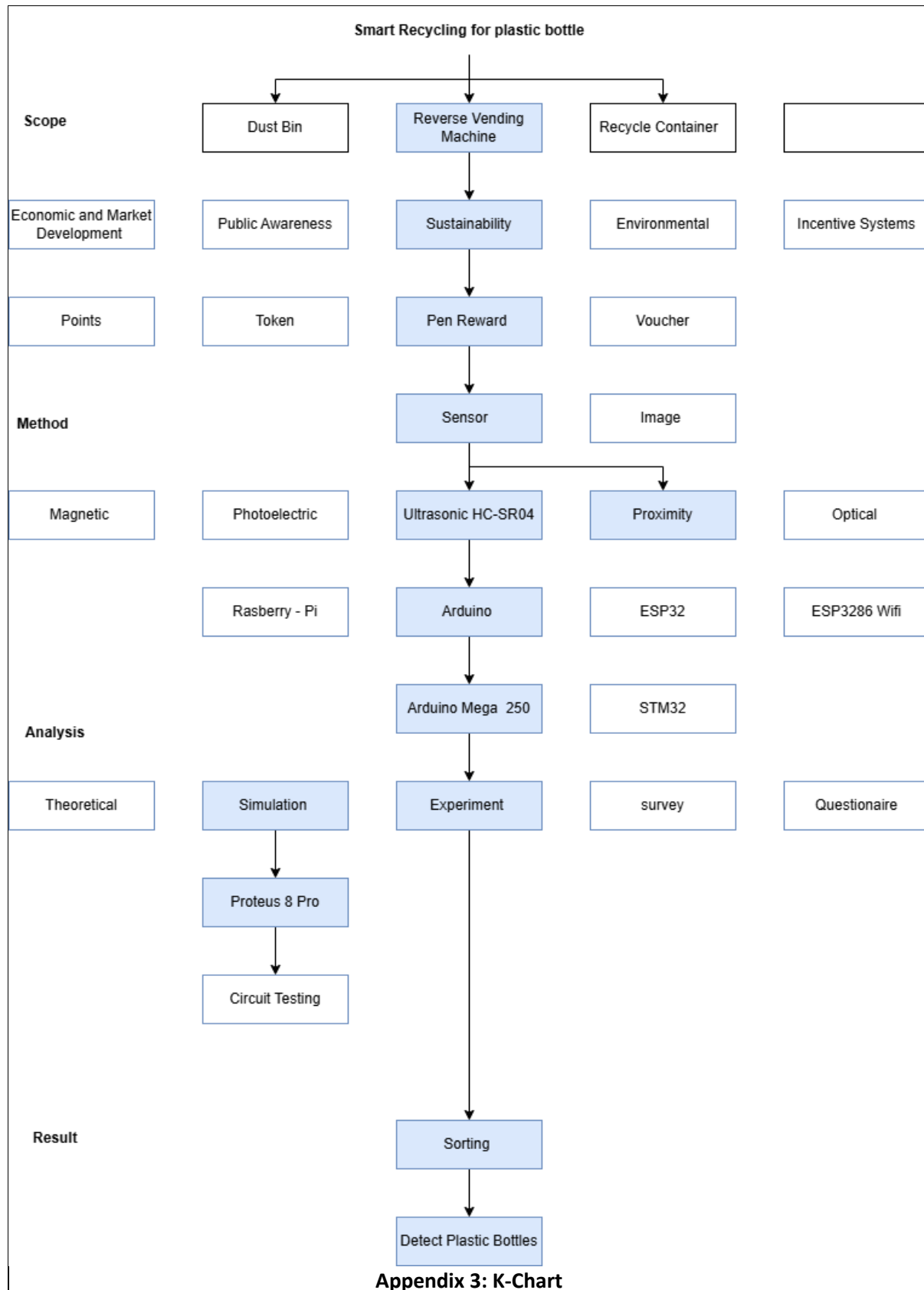
5Pcs 12V DC Power Connector Plug Jack Male ...

LJ12A3-4-Z/BX Inductive Proximity Sensor Swit.

10/20/50/100pcs DC Power Jack Female Socke...

Appendix 1: Ordered Items List







Comments (by Supervisor)	Signature and Name
ok, next progress 2	